

Coexistence

Today's Agenda:

- Quiz
- Coexistence

Paradox of the plankton

- Species need different niches to coexist
- How can so many species of plankton coexist?
 - All need the same resources
 - Well-mixed, homogeneous environment

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THE PARADOX OF THE PLANKTON*

G. E. HUTCHINSON

Osborn Zoological Laboratory, New Haven, Connecticut

Paradox of the plankton

- Species need different niches to coexist
- How can so many species of plankton coexist?
 - All need the same resources
 - Well-mixed, homogeneous environment

Or, more generally, how can so many species coexist in a given location when they all need more or less the same things (oxygen, water, carbon, etc.)?

THE PARADOX OF THE PLANKTON*

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How to coexist

We now know the answer (in a broad sense)

What needs to happen for species to coexist?

How to coexist

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What needs to happen for species to coexist?

$$\alpha_{ii} > \alpha_{ji}$$

$$\alpha_{jj} > \alpha_{ij}$$

How to coexist

What causes species to limit themselves more than their competitors?

- Different resources
- Different predators

How to coexist

What causes species to limit themselves more than their competitors?

- Different resources
- Different predators
- Spatial variation (use different locations)
- Temporal variation (use different times)
- Some combination of these things

Coexistence via spatial variation

- Variation in resources (or predators) within a patch
- Different species do best with different levels of that resource/predator

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Trade-offs

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Trade-offs

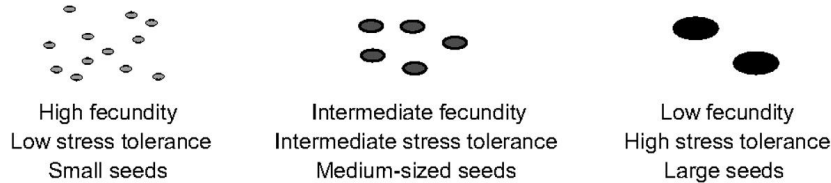
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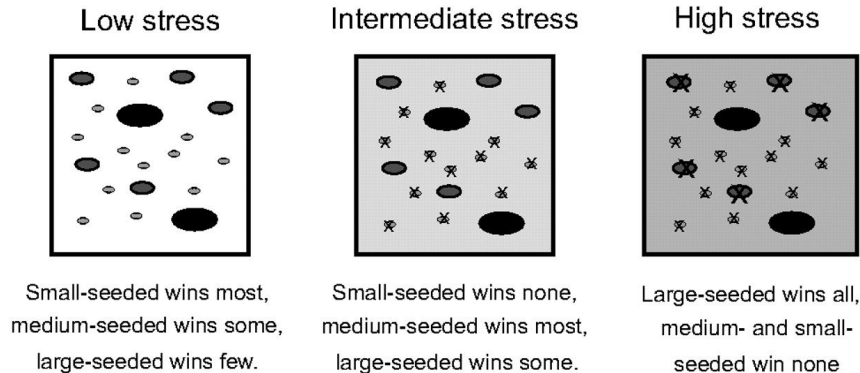
Tolerance-Fecundity

The tolerance-fecundity tradeoff

A Species vary inversely in fecundity and stress-tolerance



B Regeneration sites vary in stressfulness



The tolerance–fecundity trade-off and the maintenance of diversity in seed size

Helene C. Muller-Landau¹

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Edited* by Simon A. Levin, Princeton University, Princeton, NJ, and approved January 19, 2010 (received for review October 8, 2009)

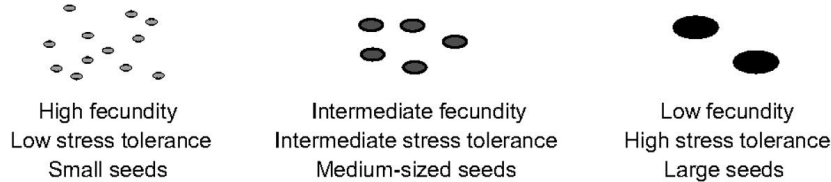
Seed size commonly varies by five to six orders of magnitude among coexisting plant species, a pattern ecologists have long sought to explain. Because seed size trades off with seed number, small-

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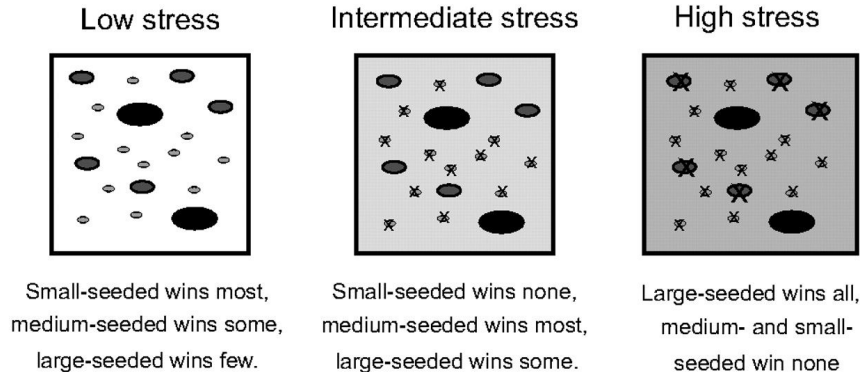
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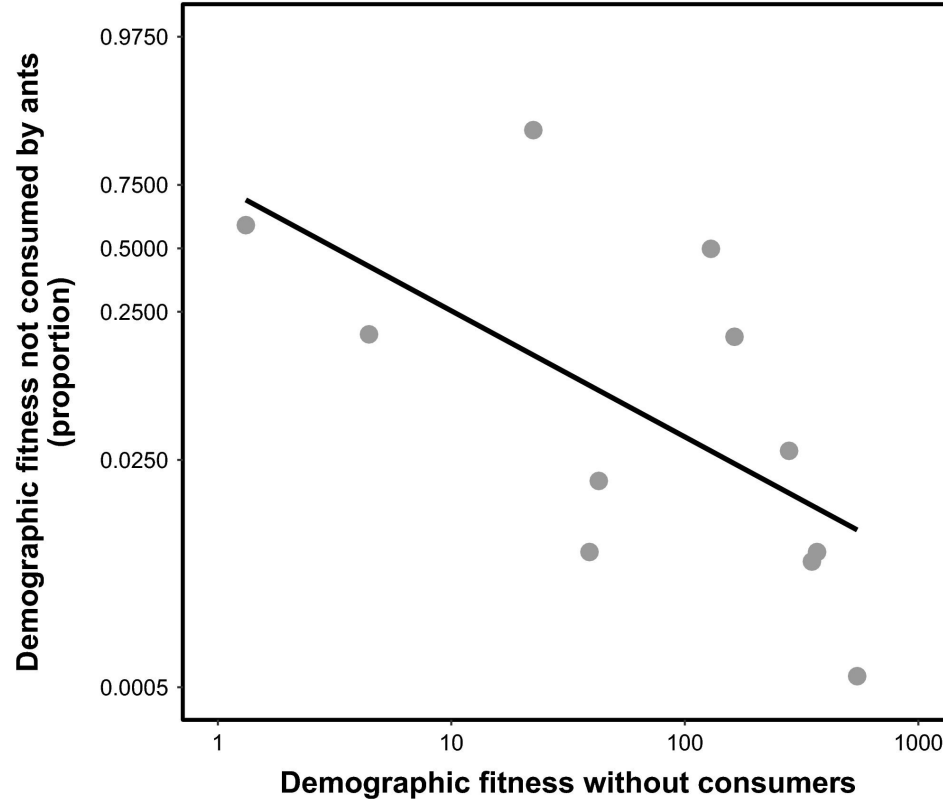
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Found theoretical support

Competition-Defense Trade-off



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BIOTIC CONTROLS OF PLANT COEXISTENCE

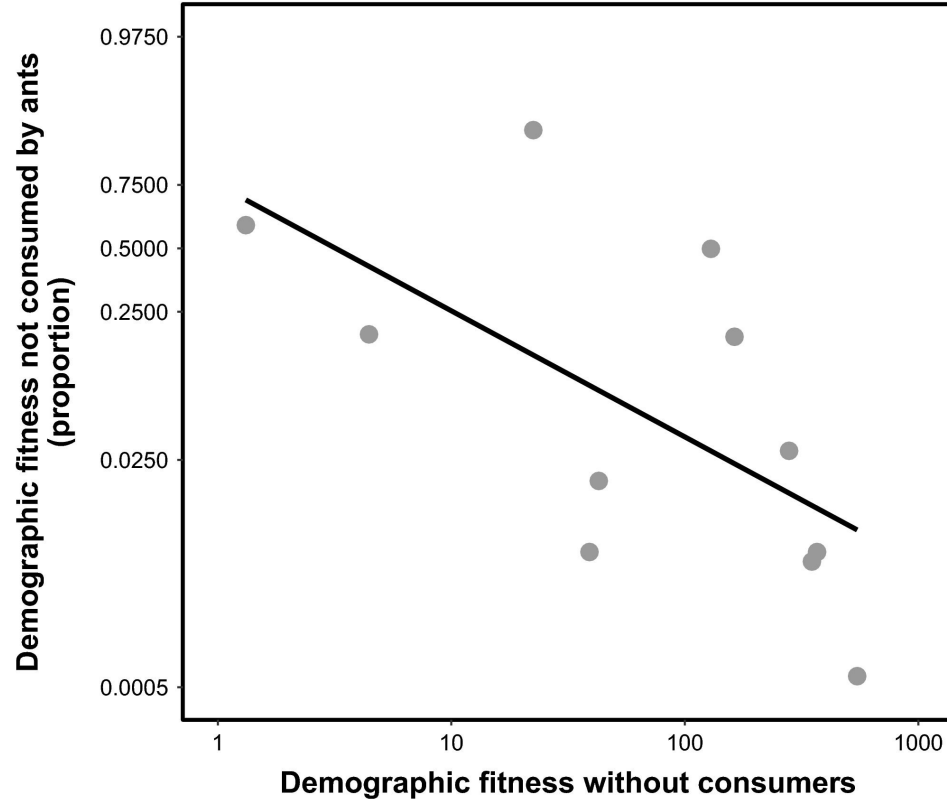
Journal of Ecology



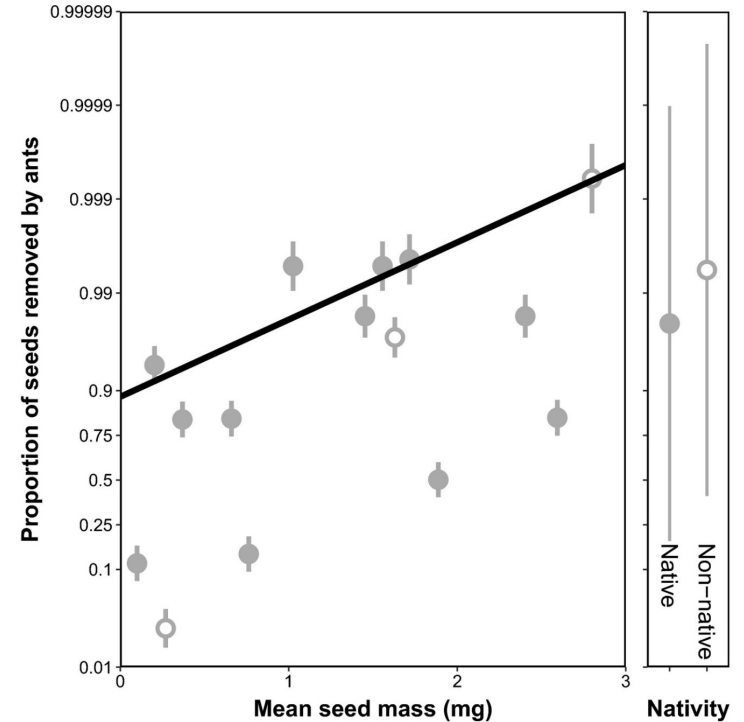
A competition–defence trade-off both promotes and weakens coexistence in an annual plant community

William K. Petry¹ | Gaurav S. Kandlikar² | Nathan J. B. Kraft² |
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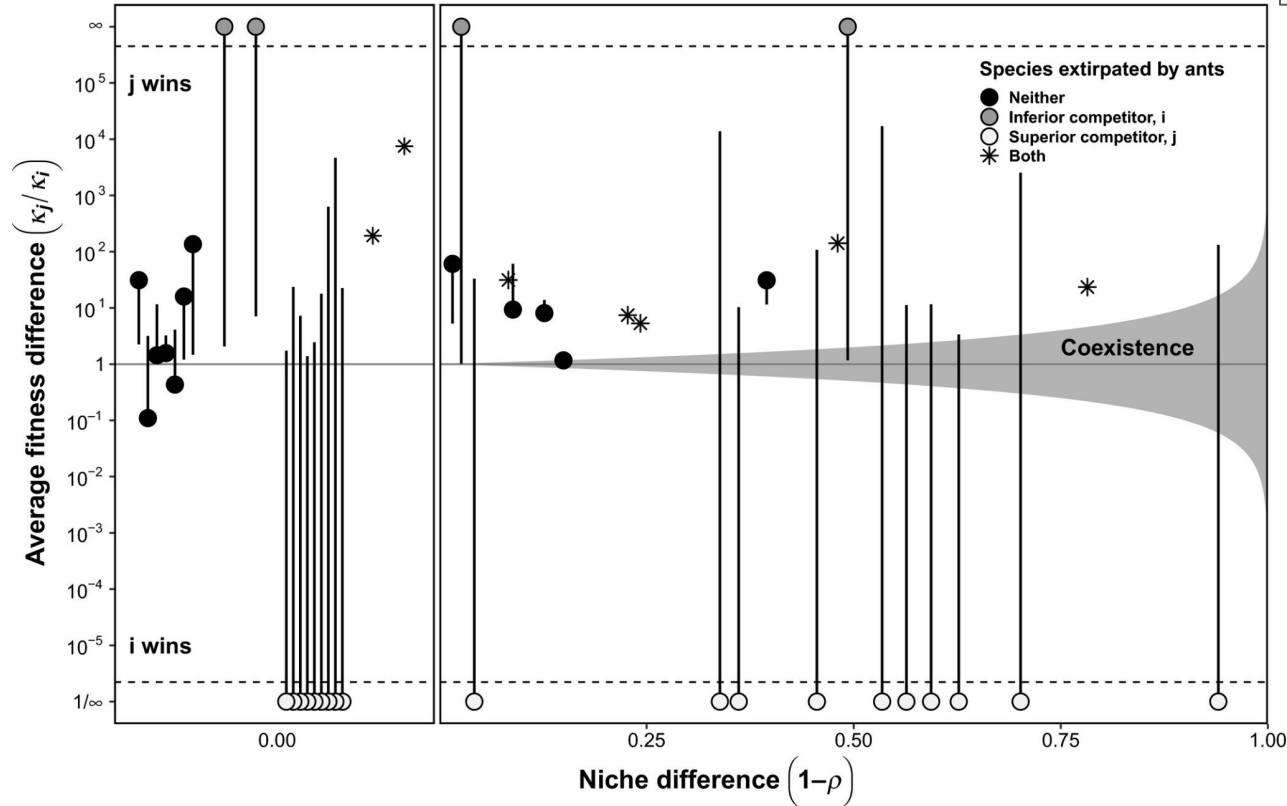
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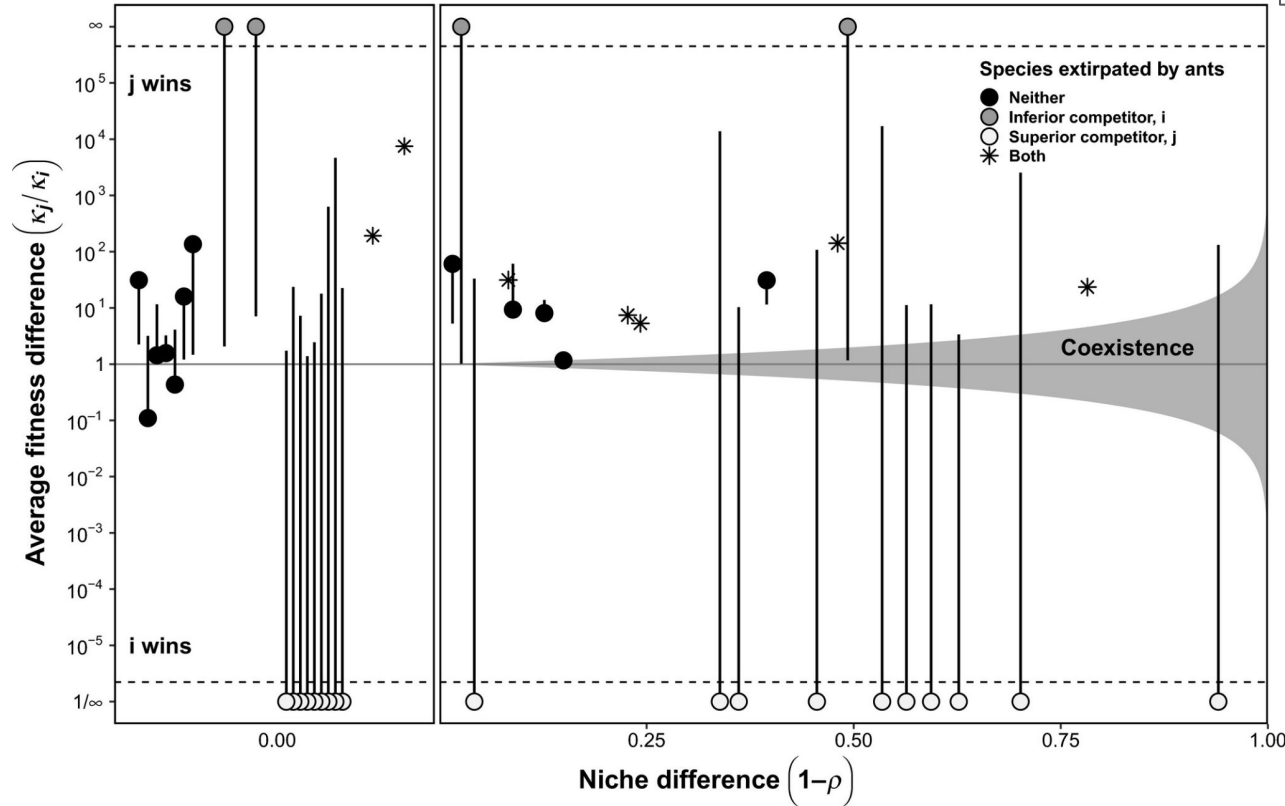
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Competition-Defense Trade-off



Found trade-offs, but...

Trade-off sometimes helped, sometimes made things worse

Coexistence without resource/predation differences

- Different preferences
- Different habitats
- Different strata within a habitat

Coexistence without resource/predation differences

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- Different habitats
- Different strata within a habitat

Same resource, but different locations: increases α_{ii} relative to α_{ji}

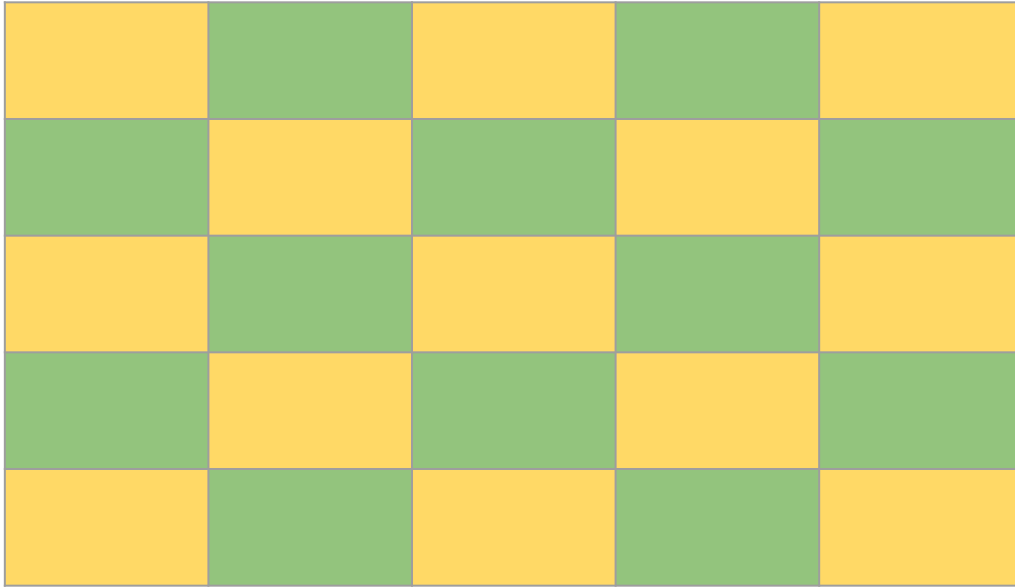
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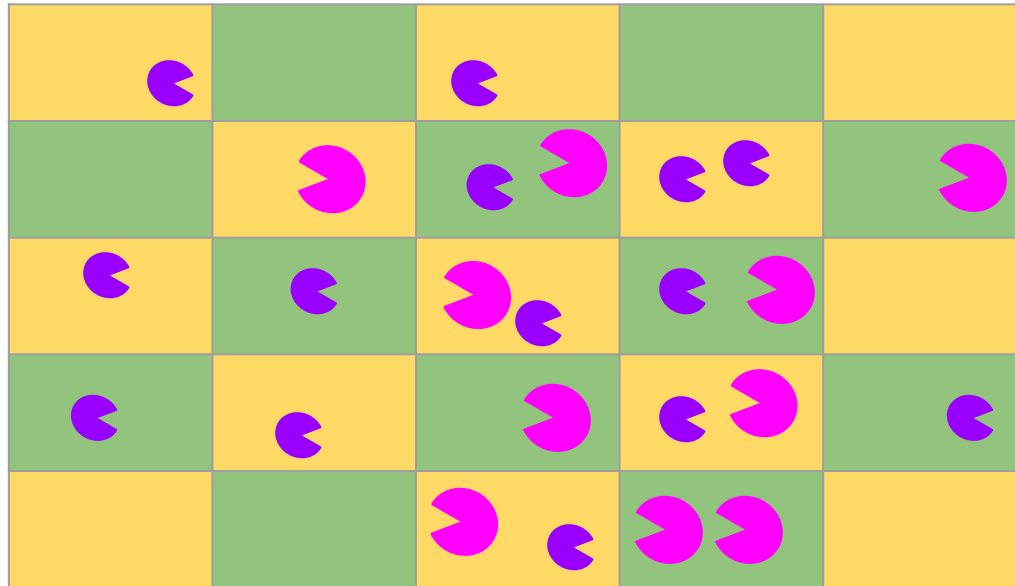
Same resource, but different locations: increases α_{ii} relative to α_{ji}

Note that there are a few caveats about this!

Coexistence without resource/predation differences



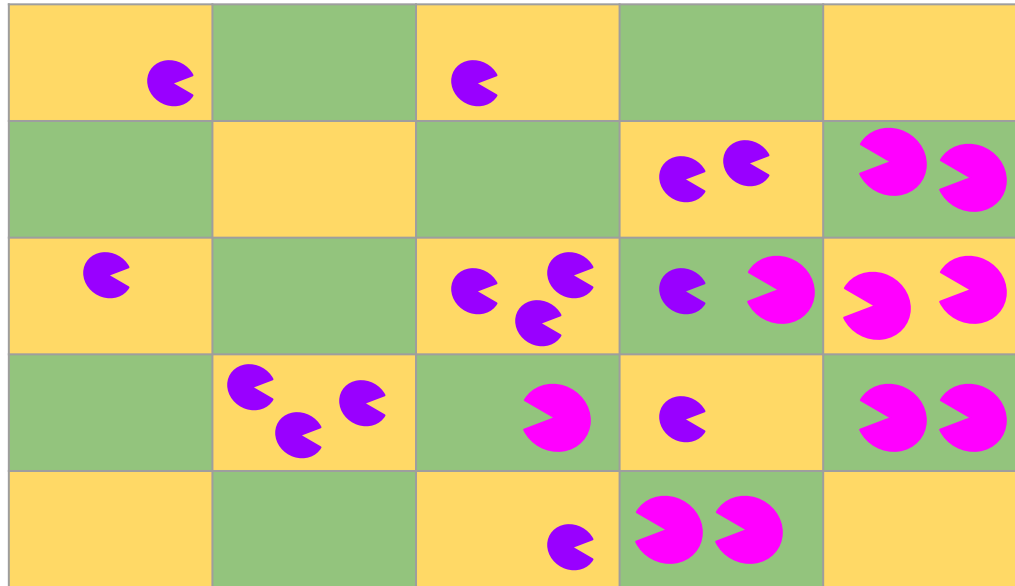
Coexistence without resource/predation differences



 = Competitor

Similar preferences

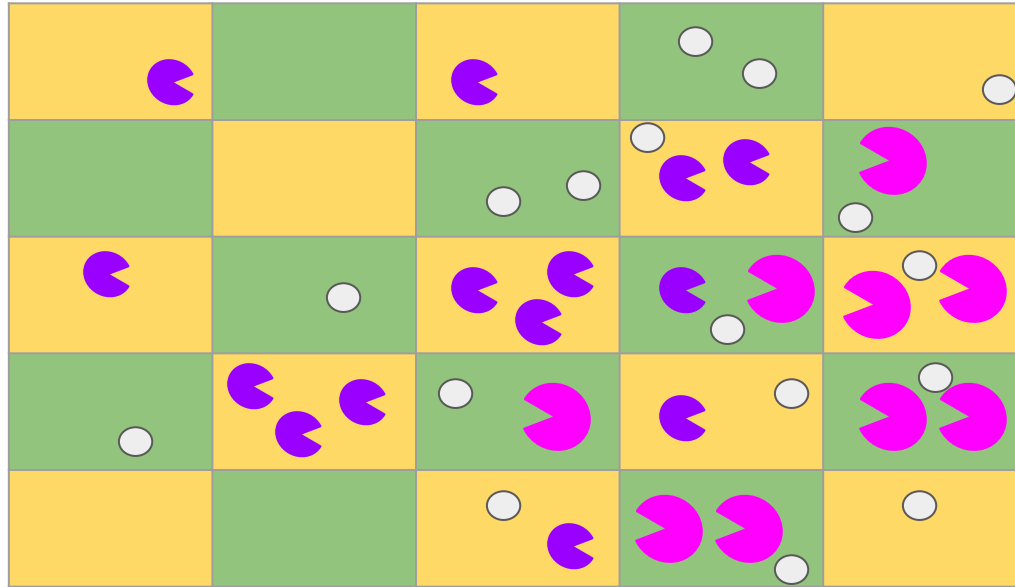
Coexistence without resource/predation differences



 = Competitor

Distinct preferences

Coexistence without resource/predation differences



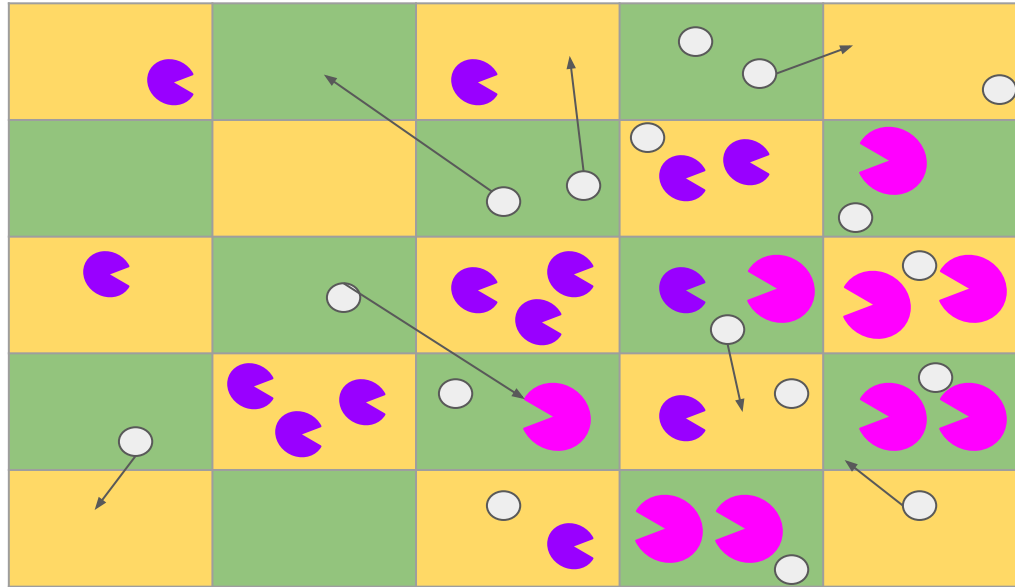
 = Competitor

 = Resources

Distinct preferences

Mostly intraspecific

Coexistence without resource/predation differences



-  = Competitor
-  = Resources
(can now move)

Distinct preferences

No longer consuming
unique resource,
increases **interspecific**
competition

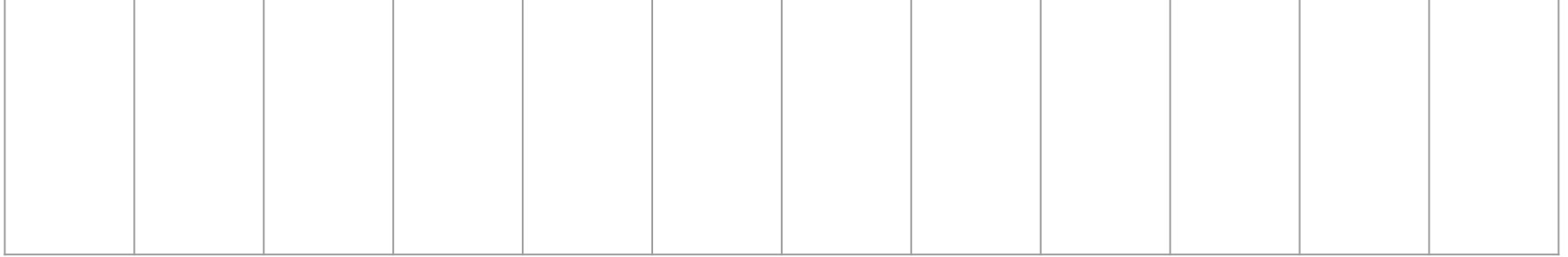
Coexistence via temporal mechanisms

- Temporal niche partitioning
- Relative non-linearity
- Storage effects

Temporal Niche Partitioning

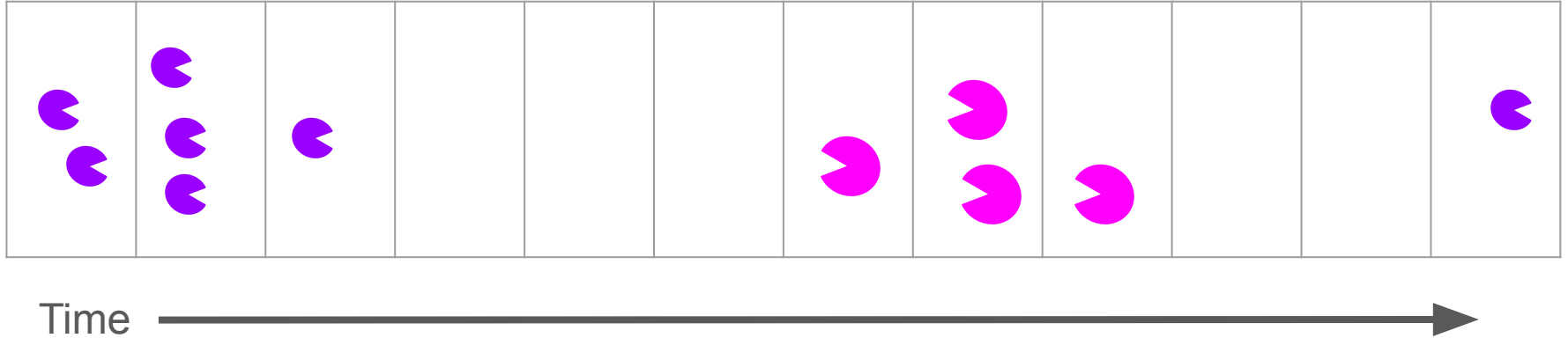
- Different species use resources at different times
- Note that the resource pools need to be different

Temporal Niche Partitioning

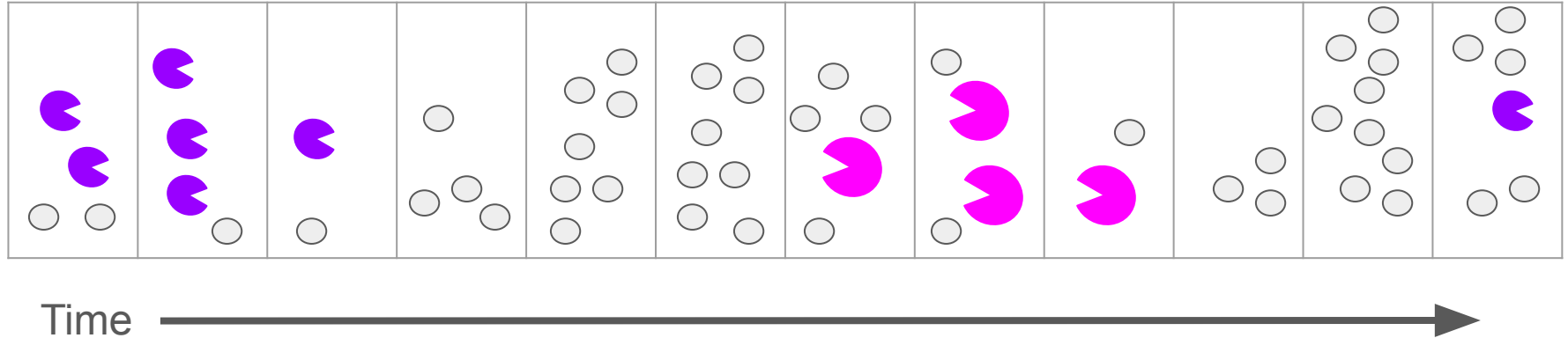


Time 

Temporal Niche Partitioning

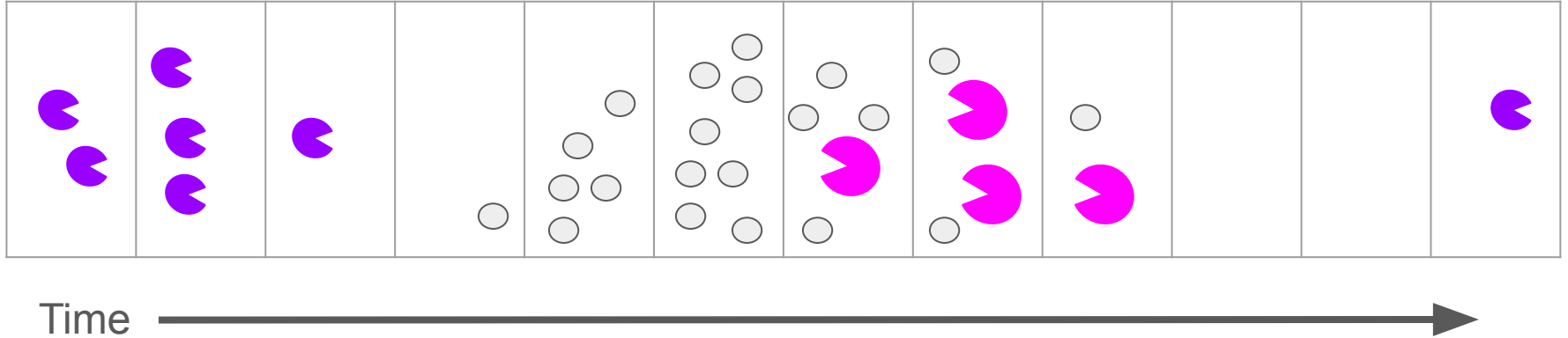


Temporal Niche Partitioning



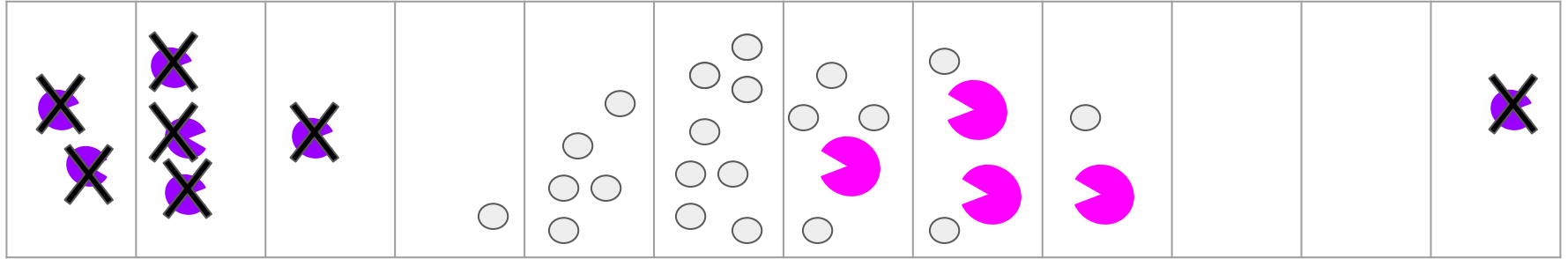
Resources are renewed over time
- can increase intraspecific competition vs interspecific

Temporal Niche Partitioning



Resources are renewed rarely
- dominance by one competitor

Temporal Niche Partitioning

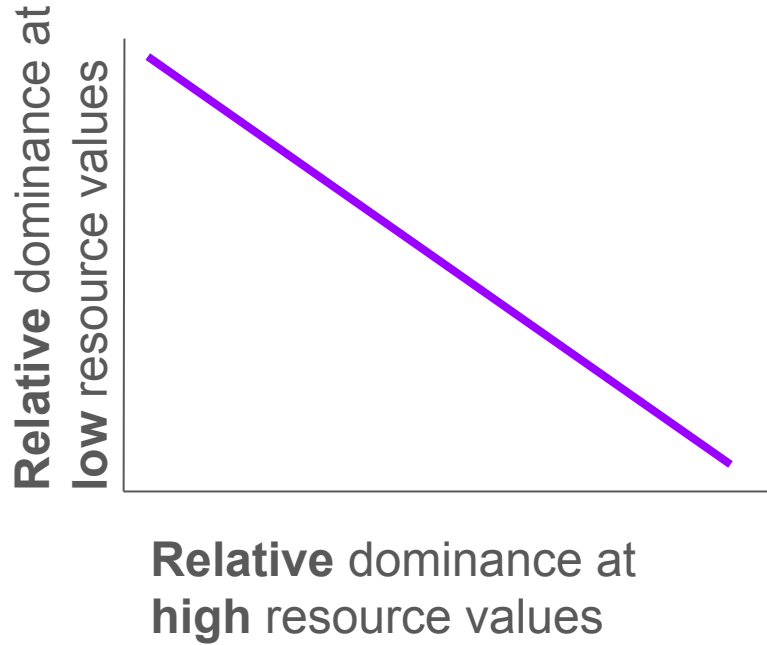


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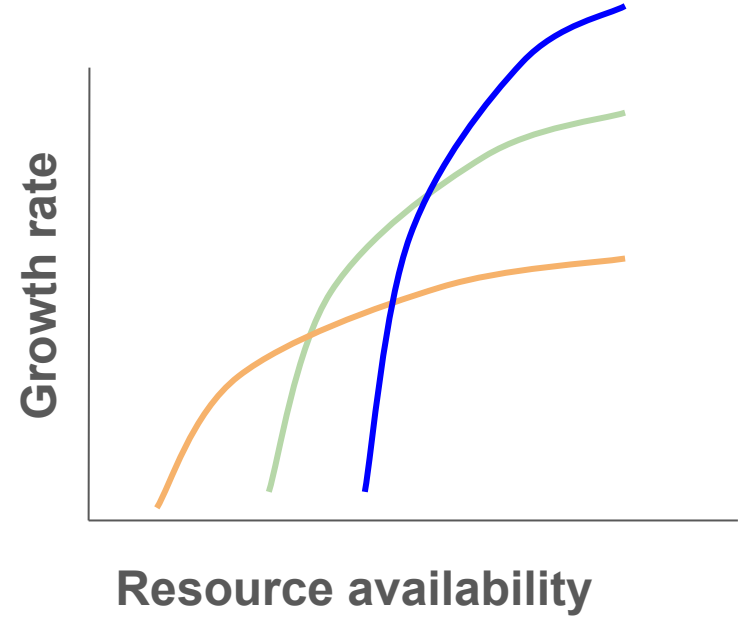
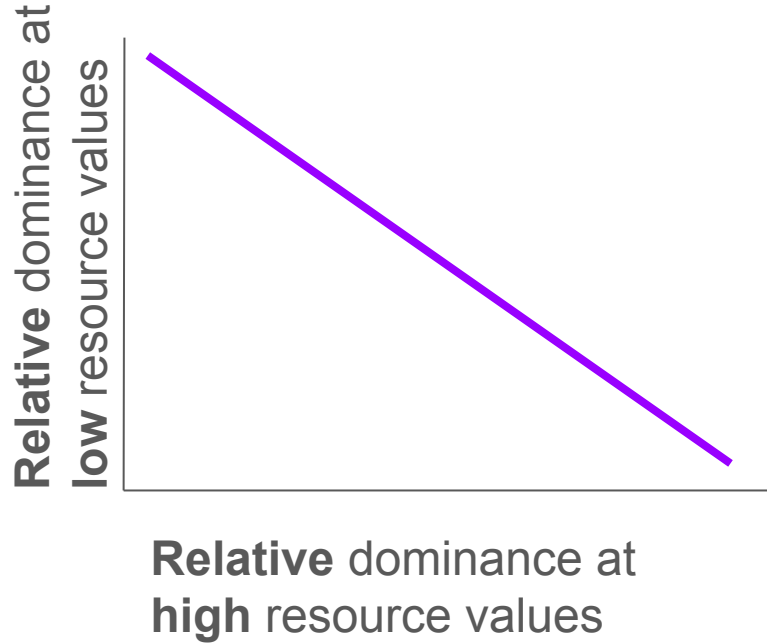
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Relative Nonlinearity of competition

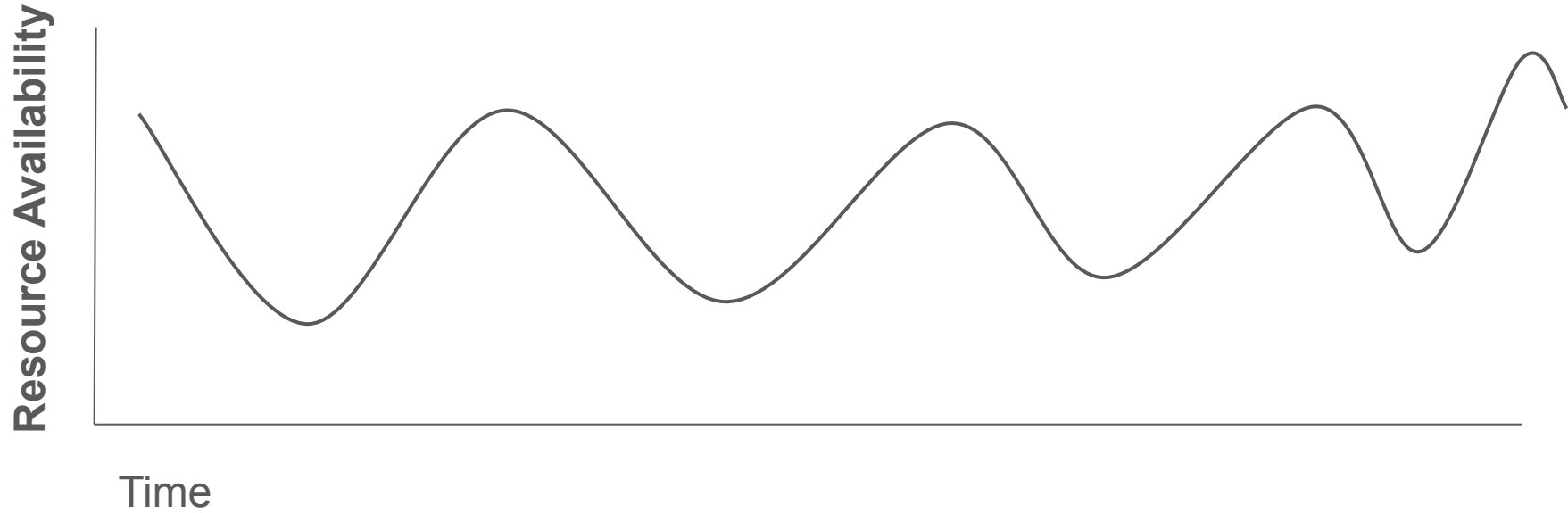
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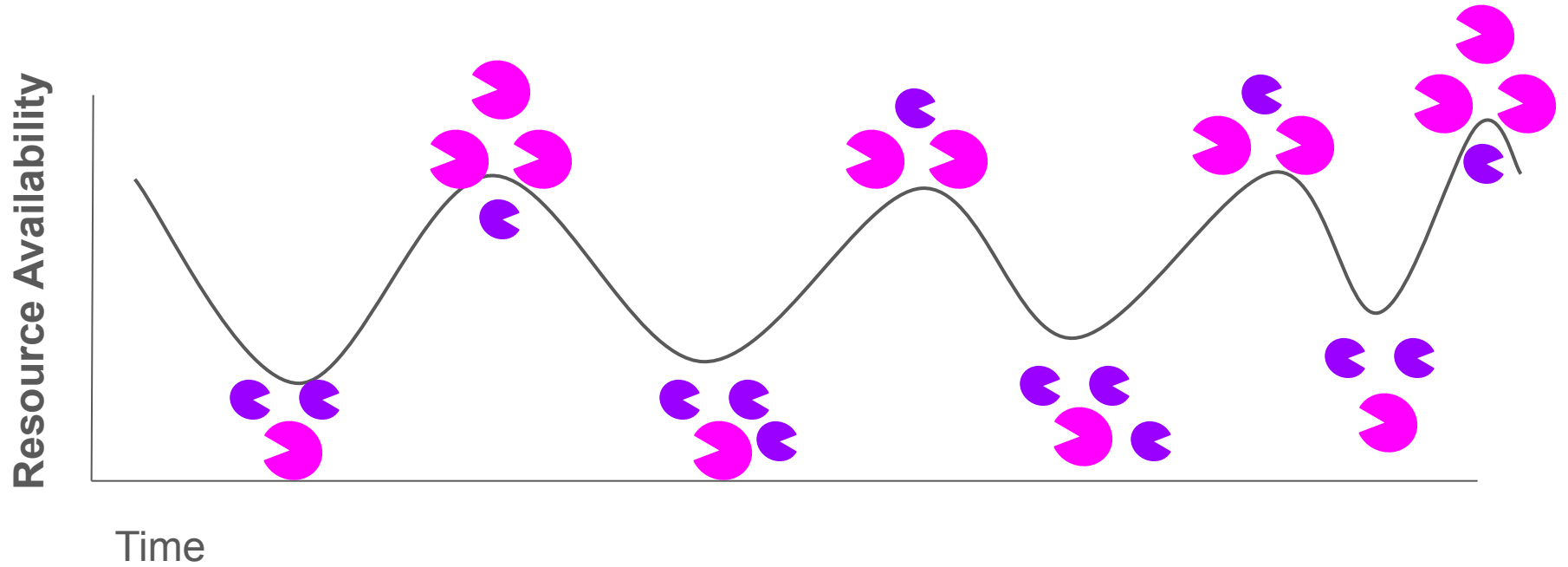
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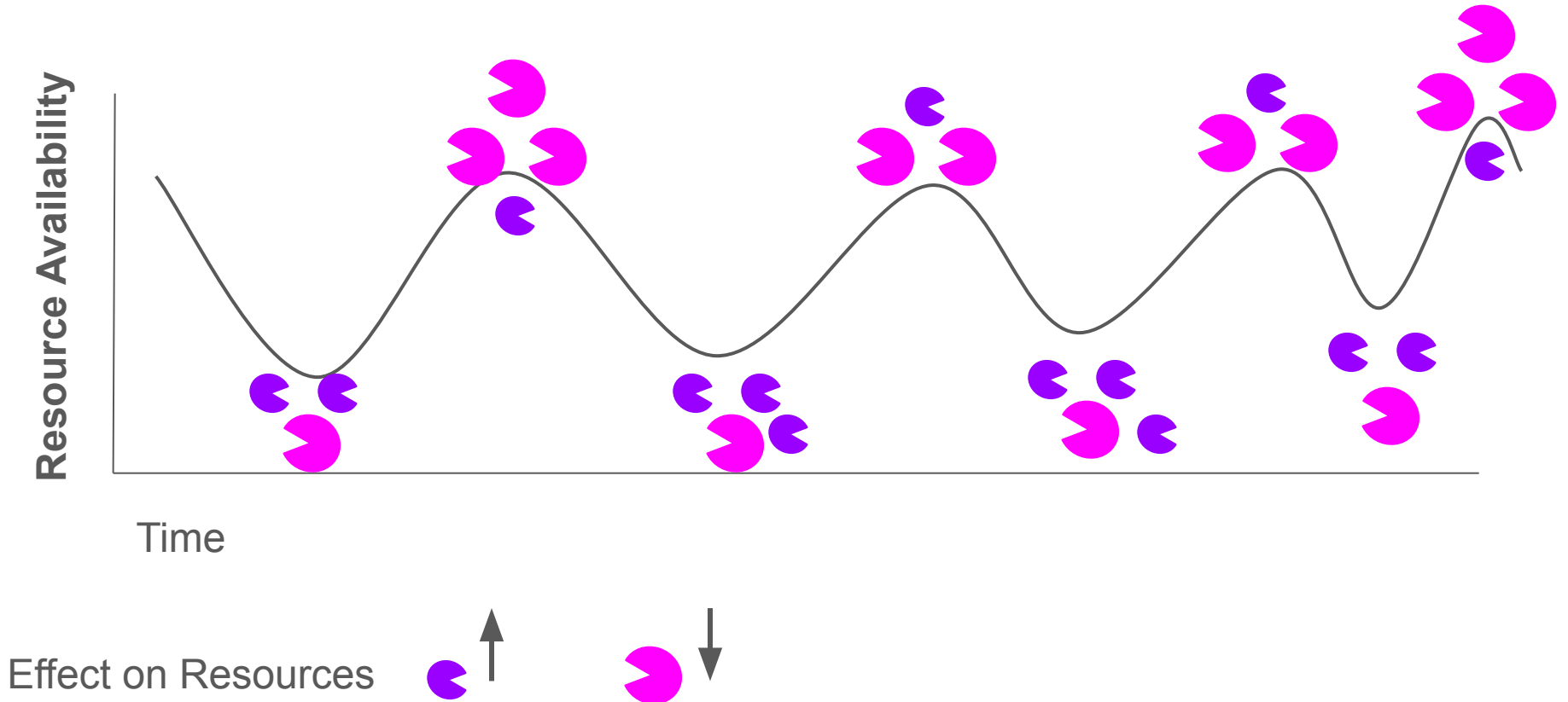
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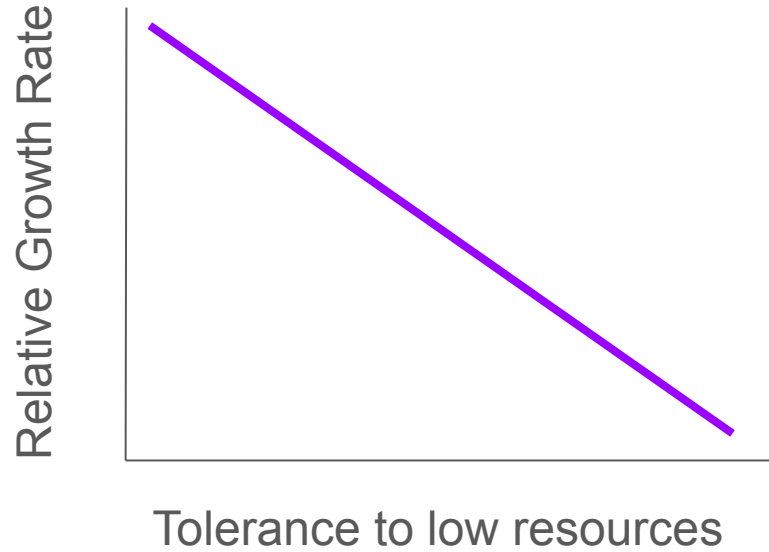


Storage Effect

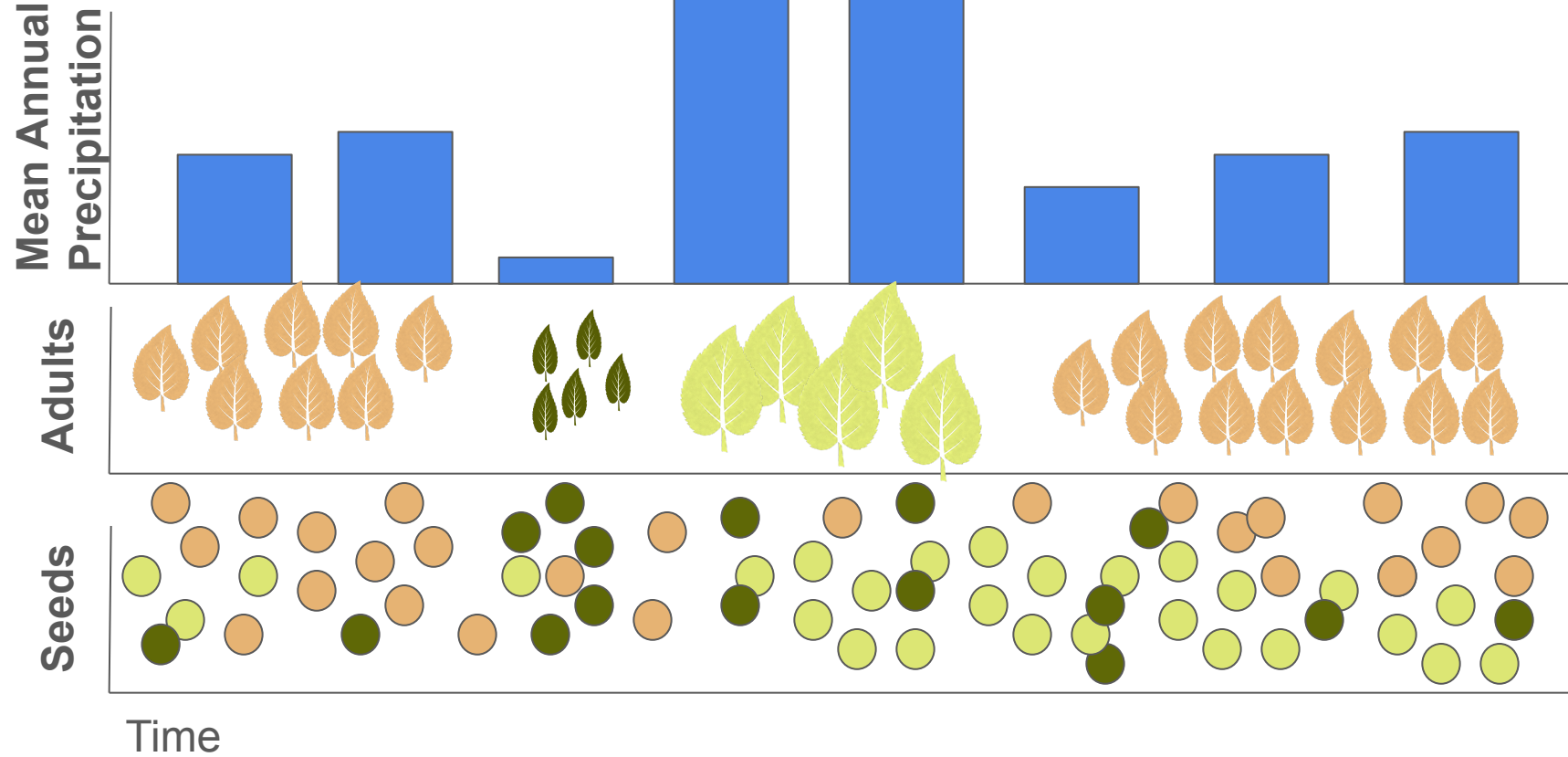
Storage Effect

- Species differ in the environment they do best in
- BUT they can persist through sub-optimal conditions
- (Note that there is a spatial version of this as well)

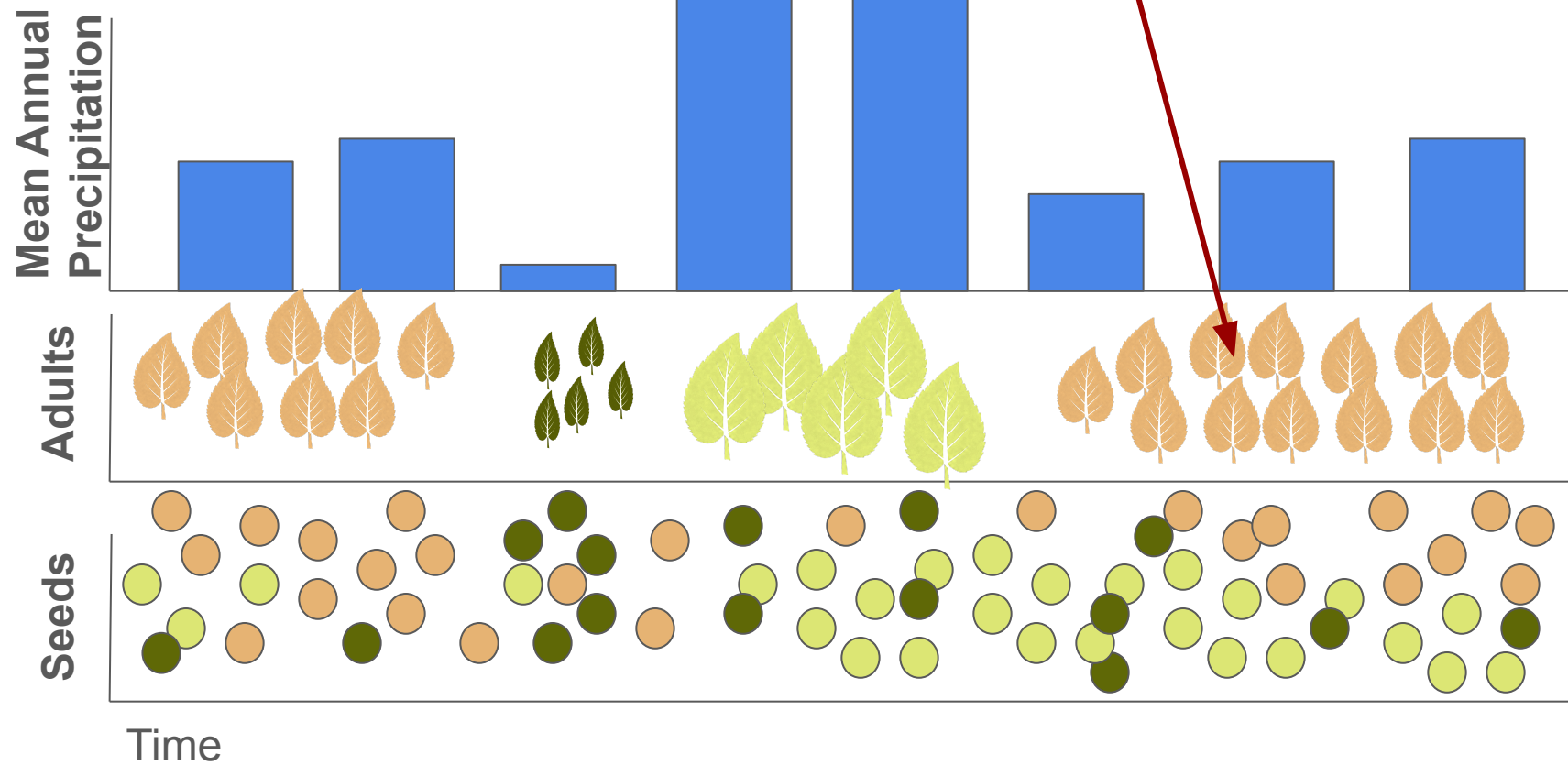
Storage Effect



Storage Effect



Storage Effect



Storage Effect

PNAS

Functional tradeoffs determine species coexistence via the storage effect

Amy L. Angert^{a,b,1}, Travis E. Huxman^{b,c}, Peter Chesson^b, and D. Lawrence Venable^b

^aDepartment of Biology, Colorado State University, Fort Collins, CO, 80523; and ^bDepartment of Ecology and Evolutionary Biology and ^cB2 Earthscience, University of Arizona, Tucson, AZ 85721

Communicated by José Sarukhán, Universidad Nacional Autónoma de México, Mexico D.F., Mexico, April 28, 2009 (received for review March 23, 2009)

How biological diversity is generated and maintained is a fundamental question in ecology. Ecologists have delineated many demographic decoupling of species arises from partially uncoupled responses to environmental variation, (i) the strength

Functional tradeoffs determine species coexistence via the storage effect

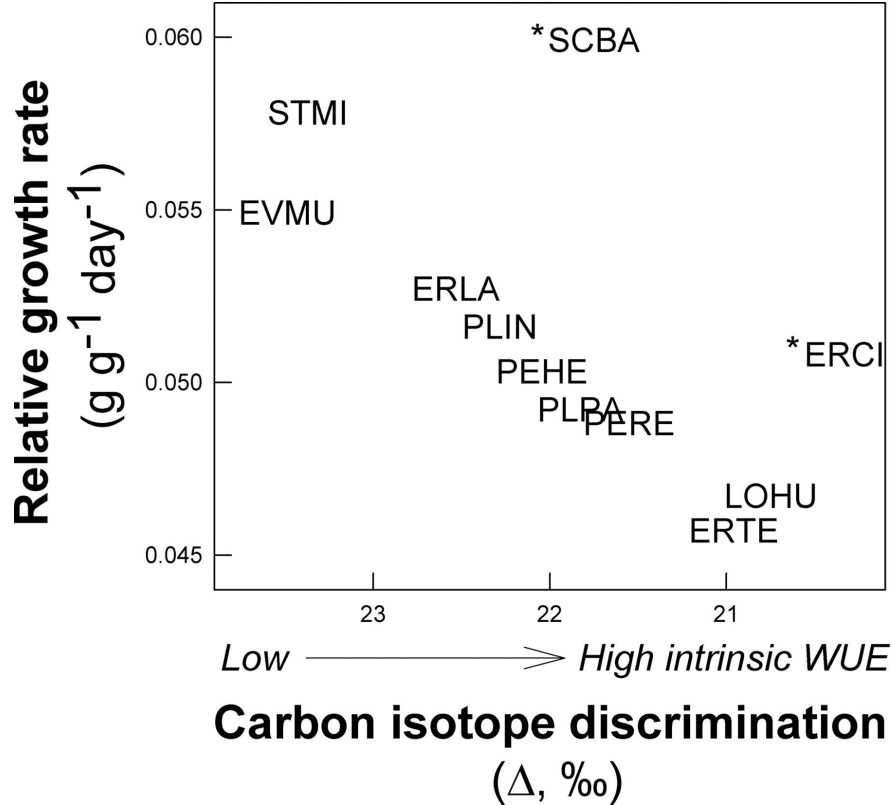
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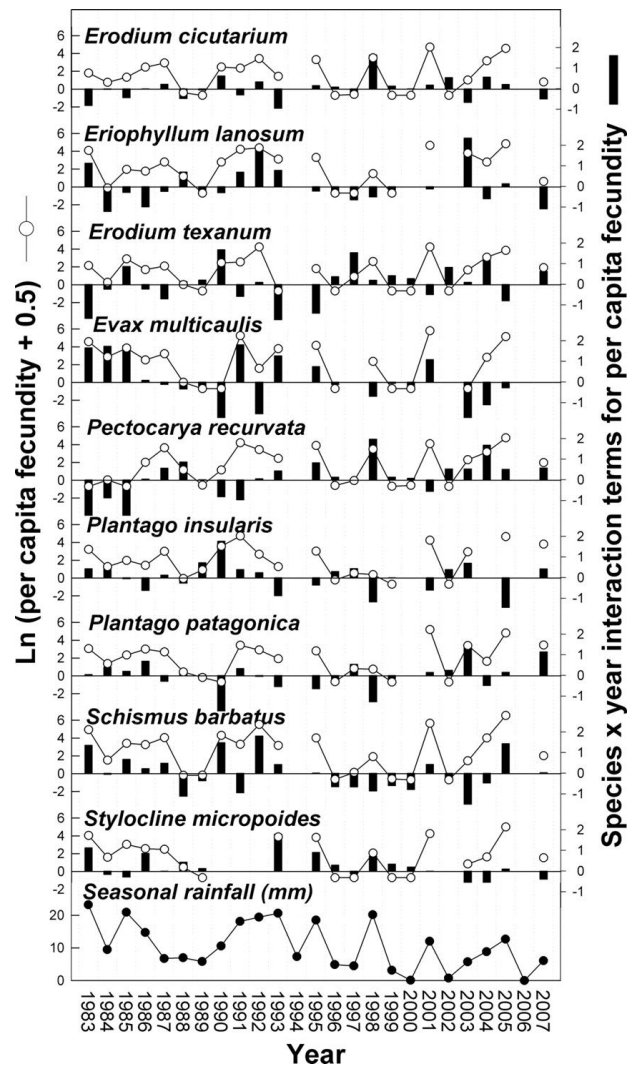
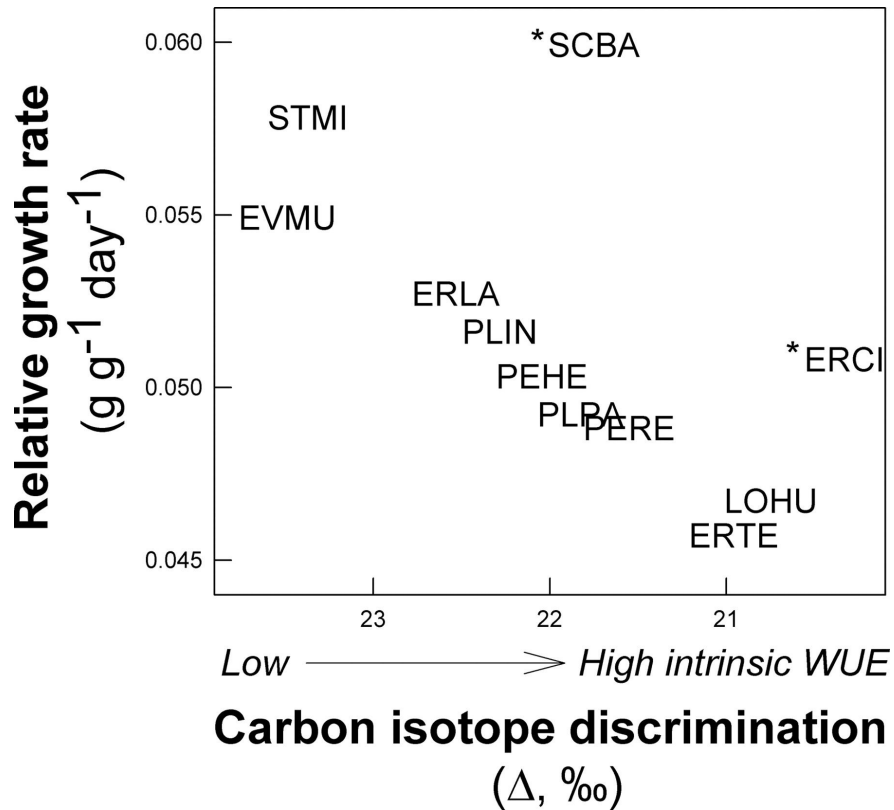
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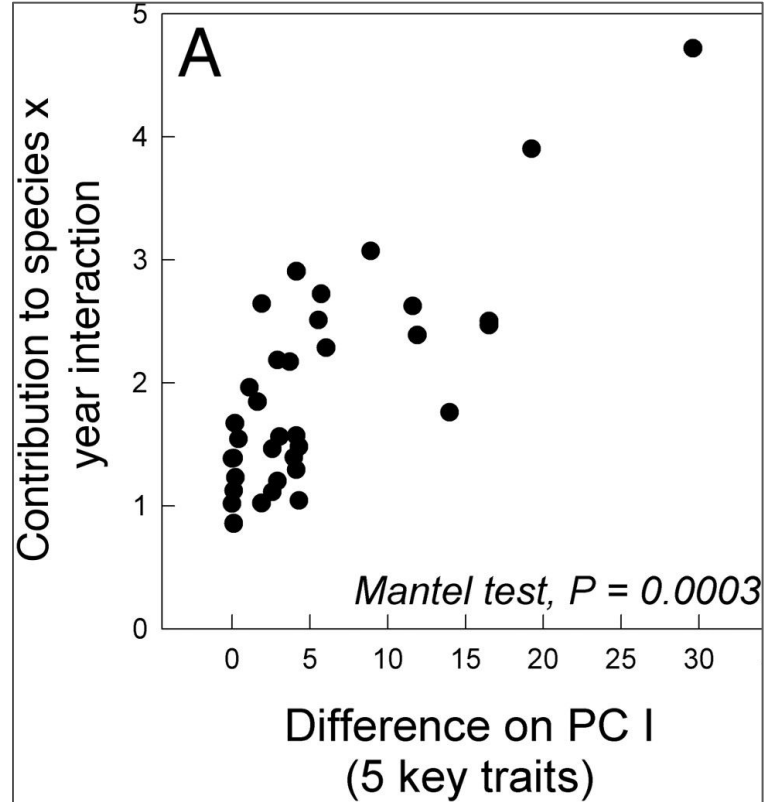
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Main Takeaways

- Species coexistence requires meaningful differences
- Species can differ in resource/predation in space and/or time
- Trade-offs are critical!
- Trade-offs need to make $\alpha_{ii} > \alpha_{ji}$ for coexistence

Next class: Mutualism and Facilitation